## Order Pholidota, Family Manidae, Genus *Manis*

Throughout Africa and across southern Asia, the semifossorial, solitary, armoured pangolins (*Manis* 2-33 kg) dig for ants and termites. Males may be larger than females (5226a, 8501, 9175). The 13 g (n1), non-deciduate, epitheliochorial or endotheliochorial, villous, diffuse placenta with interanastomosing placental strips has a superficial, antimesometrial implantation into a bicornuate uterus with endometrial glands (2391, 2396, 2397, 4552, 6936, 7563, 10310, 10958, 11185). Placentophagia has not been observed (8202). A postpartum estrus occurs 9-16 d (8200, 8202) after the birth of a well-developed, scaled neonate whose eyes open either at birth or within 9 d (31, 5758, 6936, 7206, 7421, 8202, 9175, 11162). The 2 pectoral mammae are well developed in both sexes (9175, 10158). Males are ascrotal with inguinal testes and lack bulbourethral glands (281a). Details of endocrinology (4664) are available.

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| **NEONATAL MASS** | | |
| *Manis crassicaudata* | 235 g | 31 |
| | 242 g | 7421 |
| | 400 g | 8047 |
| | 450 g | 9175 |
| *Manis gigantea* | 500 g; n2 | 5758 |
| *Manis pentadactyla* | 92.5 g; r92-93; n2 | 4556 |
| *Manis temminckii* | 120 g; n1 est premature | 5226a |
| | 256 g; n1 | 6180 |
| | 336-420 g | 11162 |
| | 340 g | 5758 |
| *Manis tricuspis* | r90-150 g; n10 | 8200, 8202 |
| | 100 g; n1 | 7206 |
| **NEONATAL SIZE** | | |
| *Manis crassicaudata* | 310 mm; 3 d | 9175 |
| *Manis gigantea* | 450 mm; n2 | 5758 |
| *Manis temminckii* | 150 mm | 5758 |
| | TL: 225 mm; n1 est premature | 5226a |
| | 290 mm | 6180 |
| *Manis tricuspis* | r300-350 mm; n10 | 8200, 8202 |
| **WEANING MASS** | | |
| *Manis gigantea* | 6 kg | 8202 |
| *Manis tetradactyla* | 400 g | 8202 |
| *Manis tricuspis* | 600 g | 8202 |
| | 750 g; n1 | 7206 |
| **LITTER SIZE** | | |
| *Manis crassicaudata* | 1 yg; n1 lt | 8720 |
| | mode 1; r1-2 | 8497, 8501 |
| *Manis javanica* | 1 | 11185 |
| | 1 yg | 556, 557 |
| *Manis pentadactyla* | 1 yg ea; n2 f | 4556 |
| *Manis temminckii* | 1 | 10158 |
| | mode 1 yg; r1-2 | 5226a |
| *Manis tetradactyla* | 1.05; r1-2; n20 lt | 8199 |
| *Manis tricuspis* | 1 | 8202 |
| | 1 | 5277 |
| | 1 emb; n1 f | 2953 |
| | 1 emb; n1 f | 8854 |
| | 1; n1 | 7206 |
| **SEXUAL MATURITY** | | |
| *Manis tricuspis* | 2 yr | 8200 |
| **CYCLE LENGTH** | | |
| *Manis tricuspis* | 9-14 d | 8200 |
| | 9.3 d; mode 4-8 62%; r3-29; n36 cycles, 4 f | 10621 |
| **GESTATION** | | |
| *Manis crassicaudata* | 65-70 d | 9175 |
| *Manis temminckii* | 139 d | 11162 |
| | 139 d | 8202 |
| *Manis tricuspis* | est ~4 mo | 5758 |
| | >5 mo; n1; capture to birth | 8200, 8202 |

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| **LACTATION** | | |
| *Manis pentadactyla* | milk production: 86 d | 6936 |
| *Manis temminckii* | solid food: 1 mo | 5758 |
| *Manis tetradactyla* | nurse: until next gestation | 8199 |
| *Manis tricuspis* | solid food: 3 mo | 8202 |
| | wean: 3-4 mo | 8202 |
| | wean: 7.5 mo; n1 | 7206 |
| **INTERLITTER INTERVAL** | | |
| *Manis tricuspis* | 0.5 yr | 8202 |
| **SEASONALITY** | | |
| *Manis crassicaudata* | preg f: July; n1; Sri Lanka | 8501 |
| *Manis gigantea* | birth: end Sept-Oct; Uganda | 5758 |
| *Manis javanica* | birth: Feb-Mar, Aug, Oct; n5; Borneo, Indonesia | 556 |
| *Manis temminckii* | preg f: July; S Africa | 10158 |
| *Manis tetradactyla* | lact f: Jan; n1; Nigeria | 4354a |
| | repro: est yr rd; few data; Gabon | 8199 |
| *Manis tricuspis* | testes: large yr rd; Gabon | 8202 |
| | mate: est Dec-Feb; British Camerouns | 9468 |
| | preg f: Mar; n1; Lake Kivu, Zaire | 8854 |
| | preg f: yr rd, not equally distributed; Gabon | 8200 |
| | birth: peak est Nov-Feb; few data; Uganda | 5758 |
| | repro: est yr rd; few data; Gabon | 8199 |